Gokulakannan Sakthivel

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Education

- **University of Maryland College Park (College Park, MD)** Sep. 2025 – May 2027 (Expected)
  Master of Science in Applied Machine Learning; GPA: 3.83
- **National Institute of Technology Rourkela (Rourkela, India)** Jun. 2018 – Jun. 2022
  Bachelor of Technology in Chemical Engineering; GPA: 3.56 (8.17/10.0)

Technical Skills

- **LLM Applications & Agents**: Production LLM applications, Agents, LLMs, AI Coding Agents (Claude Code, daily engineering use), LangChain, OpenAI, Gemini, PyTorch, HuggingFace, Python, FastAPI
- **Retrieval & Evaluation**: RAG over PDF/PPT/Word/Excel, Knowledge Graphs, OpenSearch, embeddings, answer-grounding validation, golden-dataset validation, LLM eval harnesses (30+ responsible-AI metrics)
- **Cloud, Security & Delivery**: AWS (ECS, Lambda, SQS, S3, SageMaker), Azure (Bicep, AKS, Functions, OpenAI), Terraform, Docker, GitHub Actions, IaC, CI/CD, DevSecOps, Git

Experience

- **Institute for Physical Science and Technology (UMD, MD) - Research Engineer** *Oct 2025 - Present*
  - As ***Lead Engineer***, **built the dashboard and architected the frontend** of an **RNA structure-prediction platform**, and shipped it end-to-end from job submission to results delivery using **Claude Code as a daily agentic coding assistant across the stack**, drawing **200+ users and 600+ folding jobs in its first week**.
  - Migrated folding workloads off a shared university HPC queue onto a dedicated **SLURM cluster on AWS** with an asynchronous **SQS dispatch pipeline**, running **20 jobs concurrently** and eliminating queue wait.
  - Architected the full AWS estate as **IaC** (**Terraform**: ECS Fargate, Lambda, SQS, S3) with a **Dockerized CI/CD pipeline** (GitHub Actions → ECR → ECS).
  - Advised by Prof Pratyush Tiwary, training **RNA language models from scratch** on unsupervised Masked Language Modeling (MLM) objectives; **co-developed *REDIAL*** to score model quality **without labeled structures**, exposing over-parameterization in existing models.
  - **Training and finetuning RNA language models** to give **more accurate contact maps** and **better embedding space**, so that **downstream tasks are easier to work upon**.
- **Centre for Responsible AI (IIT Madras, India) - Applied AI Engineer** *Apr 2025 - Aug 2025*
  - ***Founding Engineer*** of a unified **LLM evaluation suite** consolidating **30+ responsible-AI metrics** (**hallucination, toxicity, bias, adversarial robustness, and multilingual accuracy across Indian languages**) from scattered open-source libraries into one modular harness.
  - Benchmarked **ChatGPT, Sarvam**, and other Indian-market chatbots across **100+ research-curated testcases**.
- **ThirdAIAutomations (Remote, India) - Applied AI Engineer** *Jan 2025 - Mar 2025*
  - As ***Lead AI Engineer***, built a client-facing RAG and Knowledge Graph pipeline over **500+ documents across PDF, PPT, Word, and Excel** using LangChain, OpenSearch, and EC2, reaching **90% answer accuracy** against an internal golden question set; establishing the startup's first scalable document intelligence stack.
- **Eli Lilly and Company (Bengaluru, India) - Senior ML Engineer** *Jul 2022 - Jan 2025*
  - Led the Lilly side of a production **RAG chatbot** built with **Microsoft** engineers, serving **400+ internal users at 99%+ reliability**, with releases gated on a **golden-dataset validation pipeline** at **86 to 93% accuracy**.
  - Designed production **RAG architectures** on AWS, LangChain, and OpenAI across **7 critical AI projects**, with a **custom accuracy pipeline** validating answer grounding against source documents; replaced a legacy chatbot with an AI solution adopted **organization-wide**.
  - Led a three-person team to build an internal RAG chatbot (GPT4All, open-source embeddings), earning **runner-up among 40 teams** and a direct recommendation to the GenAI team.
  - Joined as an **Azure Cloud Engineer**, migrating **Azure Resource Manager (ARM)** templates to policy-compliant reusable Bicep templates and building **DevSecOps** automation around Azure OpenAI integrations.

Projects

- **MoE Ceilings**: Showed three **MoE inference optimizations don't pay**: across **OLMoE-1B-7B, DeepSeek-V2-Lite, and Qwen3-30B-A3B**, routing explains only **2.7%** of INT4 quality loss, **Triton kernels** hit a **1.07× Amdahl ceiling**, and **torch.compile** ran **3× slower than eager**.

Publications and Patents

D. Teng, Y. Qiu, **G. Sakthivel**, et al., "Detecting and quantifying overparametrization in RNA language models with REDIAL". *Under review, Proceedings of the National Academy of Sciences (PNAS)*, 2026. Preprint: bioRxiv.

D. Teng, Y. Qiu, **G. Sakthivel**, A. Aranganathan, L. Herron, and P. Tiwary, "RNAnneal-Light: Language-model based RNA ensemble structure prediction". *Invention Disclosure* (Pending), 2026.